Answer **all** questions.

Question 1: Changes in the market for seafood

Extract 1: Higher cod prices hit Europe's fish consumers

Cod is a sea fish and is the second-most-consumed fish in Europe, behind tuna in first place and ahead of salmon in third. However, Europe's consumers are facing higher cod prices (see Figure 1). This is because of increased demand for Norway's cod caused by the rising price of salmon, and a prolonged disagreement in Iceland between fishing workers and boat owners. Norway and Iceland are two of Europe's major producers of cod.

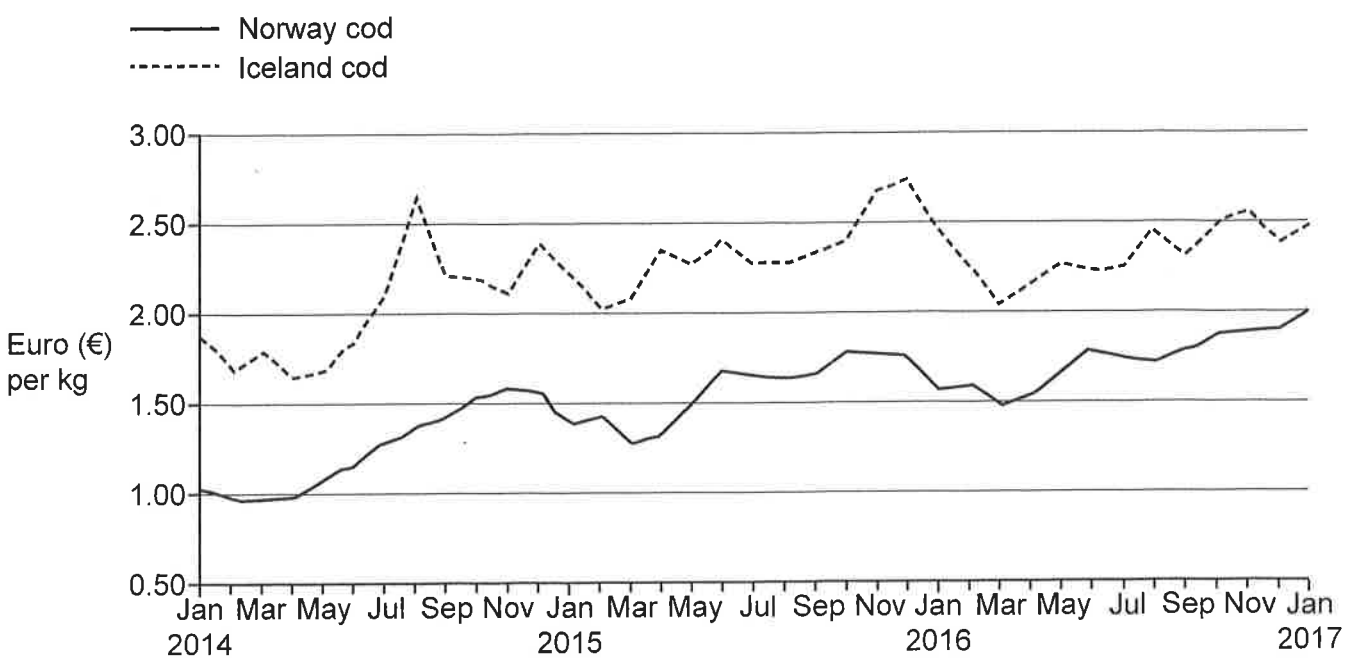
Norway's cod prices have risen by around 25 per cent from a year ago to €2.00 per kg after a 45 per cent rise in salmon prices over the same period led consumers to look for alternative fish.

In Iceland, disagreements between fishing workers and the boat owners have left fishing boats idle in the ports for over two months.

An analyst said that large stockpiles of frozen cod in Iceland have been able to keep prices low for now despite the fall in output, but these stockpiles had started to reduce. He said, 'The peak cod-fishing season in Iceland is typically between February to April, meaning current delays in fishing could reduce supplies and affect prices later in the year.'

The disagreements in Iceland are mostly over the wages for fishermen and fishing industry taxes. The Icelandic government has said it does not want to get involved, but the stalemate in talks means it may be forced to intervene.

Figure 1: European cod prices (Norway and Iceland), January 2014 to January 2017



Source: The Financial Times, 16 February 2017

Extract 2: Shrinking prospects for shrimp farmers as India faces a global slump

The cascading effects of the Covid-19 (coronavirus) pandemic in 2020 have had a striking impact on shrimp trade in India – farmers of the shellfish have been losing their livelihoods, and experts are expecting more disruption to come.

Around 80 per cent of all farmers in the village of Baguran Jalpai, which is close to the Bay of Bengal, have switched to shrimp farming recently because of relatively high revenues. But fears of losing all their hard-earned income at a single stroke have been growing with the spread of Covid-19.

Sunil Mahata, a shrimp farmer who lives in the village, expects heavy losses if the situation doesn't change in the next few months. 'I invested around US$6500 for 200 000 tiny shrimp just ten days before India's lockdown was announced. I have to spend money on feeding the shrimp and also on medicines for my family.'

An industry expert said, 'The cold storages were full of seafood and the small-scale shrimp farmers were forced to sell their product at 10 per cent of the normal price to get rid of it. I fear that many of them will go out of business. Large quantities of seafood were destroyed because there was no storage place left.'

India is the second largest fish producer in the world, with a total production of 13.7 million tonnes in 2018-19. The industry employs over 16 million people in the seafood business.

Indian exporters fear that major buyers like France, Italy, Spain, the United States (US) and Portugal might be forced to reduce their orders, as their economies will take a long time to recover from the damages of the pandemic.

An exporter said 'India's shrimp production will not be the same as before, as farmers have become reluctant to invest in shrimp because of the situation with the pandemic – many will leave the market. The shrimps farmed in India are mostly used in hotels and restaurants across the globe, and the whole hospitality and dining sector might take a hit if the pandemic keeps customers away. But maybe the situation won't be as bad as is being envisaged. We should wait and watch rather than becoming too negative over it.'

Source: The Fish Site, 23 April 2020

Extract 3: Barramundi Asia rolls out ambitious growth plan

Singapore-based Barramundi Asia has big plans for growth. The company farms the barramundi fish in deep seawater off Singapore's southern isles, as well as around the Islands of the Kimberley off northern Australia and a new farm set up in Brunei's waters. The company is working to scale annual production up to 50 000 tonnes of barramundi by 2030. Barramundi Asia uses hi-tech automated farming processes, harvesting when its fish get to four to five kilograms in size.

'The overwhelming growth in fish farming for the future is still here in Asia,' said the chief executive officer (CEO). 'Singapore is an international financial hub, and has proximity to our key business activities in Brunei and Australia as well as our local research and development sites.'

Currently, the company's production is around 4000 tonnes, with around 700 tonnes produced in Singapore and 3000 tonnes produced in Australia. It makes around SG$20 million in annual revenue – not enough to make it profitable yet. But that is due to the company's aggressive growth plans for the next ten years. By expanding its operations across its three farming sites, Barramundi Asia says it is on track to produce more than 11 000 tonnes of barramundi fish by 2023. And that could grow by even more if the company adds more new farms.

Because of the geographic distance between its farming operations, Barramundi Asia has found it makes both logistical and financial sense to own and operate its entire supply chain: breeding and processing the fish, managing the farms, as well as running a sales and marketing operation.

Barramundi Asia says it can produce its own vaccines and has methods of farming barramundi fish from small fry to fully-grown fish, as well as a sophisticated sales and marketing operation with online shops, home delivery and a strong brand. It also now has genetics expertise, after starting to work with Temasek Life Sciences, a non-profit organisation specialising in fish genomics and breeding programmes.

'Can we get to 50 000 tonnes? I believe so. It's not going to be easy but we're quite confident,' said the CEO. 'I do believe we have the capabilities in place to deliver on our expansion plans. I'm very high in confidence we can make it happen.'

Source: SeafoodSource, 28 January 2020

Extract 4: 'Plant-based seafood is the logical next step'

Plant-based foods are trending as consumers opt for healthy and sustainable food choices. Novish is a producer that specialises in 100% plant-based substitutes for fish. According to the company, natural aromas inspired by the plants, seaweed and algae, offer a delicious flavour. The products are 100% vegan and soy-free. Their first fish-free range of foods will be presented during an event in the Netherlands, Europe, being held in January 2020.

The company says plant-based seafood is the logical next step in view of 'overfishing' (where the stock of fish in the seas is reduced by unsustainable amounts) and other sustainability challenges associated with seafood.

'Similar to meat alternatives, our key challenge is to develop products which have the right taste and texture, and thus sufficient consumer appeal. We have launched our first series of products which have received excellent feedback. We will introduce other products that taste like uncoated salmon burgers and tuna steaks which are more challenging but certainly very feasible', the company said.

The target consumer group is people who worry about our planet and, therefore, want to switch from animal proteins to plant-based alternatives. The company expects the growth in consumption of plant-based seafood will 'move much faster than meat alternatives since consumers already know about the sustainability and health issues.'

Source FoodIngredientsFirst.com, 14 January 2020

(a) Using a demand and supply diagram, explain why the rise in the price of salmon led to the change in the price of cod in Norway. [2]

(b) With reference to Extract 1 and Figure 1:

(i) Using a diagram, explain the likely impact of the disagreement between fishing workers and boat owners in Iceland on the market for cod in 2017. [2]

(ii) Suppose a maximum price of Iceland cod of €2.25 per kg had been operating since March 2016.

Explain the impact such a maximum price would have had on the market for Iceland cod. [2]

(iii) Explain how 'large stockpiles of frozen cod in Iceland' can be used to keep cod prices stable, and identify **two** difficulties in operating such a scheme. [4]

(c) Explain why many small-scale shrimp farmers in India took the decision to shut down production in the short run and leave the market in 2020. [2]

(d) Discuss whether Barramundi Asia's plan to increase its scale of fish production is likely to benefit consumers. [8]

(e) Due to fears over climate change and a growing demand for plant-based diets, many consumers are changing from animal protein to vegan alternatives.

Discuss whether demand factors or supply factors have a greater impact on the market for fish in the long run. [10]

[Total: 30]